

STAT - 305

Exercise Chapter no-5

Show that

$$v(pst) = \frac{1}{n} (\sum w_h \sqrt{p_h Q_h C_h})$$
$$(\sum w_h \sqrt{p_h Q_h / C_h}) - \frac{1}{N} \sum w_h \text{ opt-c}$$

Solution:-

$$\begin{aligned} v(pst) &= \sum \frac{w_h^2 p_h Q_h}{n_h} (1 - f_h) \\ &= \sum \frac{w_h^2 p_h Q_h}{n_h} \left(1 - \frac{n_h}{N} \right) \\ &= \sum \frac{w_h^2 p_h Q_h}{n_h} - \sum \frac{w_h^2 p_h Q_h}{n_h} \cdot \frac{n_h}{N} \\ &= \sum \frac{w_h^2 p_h Q_h}{n_h} - \sum \frac{N^2 p_h Q_h}{N^2} \cdot \frac{1}{N} \\ &= \sum \frac{w_h^2 p_h Q_h}{n_h} - \frac{1}{N} \sum w_h p_h Q_h \end{aligned}$$

Substituting the value of optimum allocation when cost function is involved i.e

$$n_h = \frac{n w_h \sqrt{p_h Q_h / C_h}}{\sum w_h \sqrt{p_h Q_h / C_h}}$$

$$V(pst)_{opt.c} = \frac{\sum W_h^2 p_h q_h}{n W_h \sqrt{p_h q_h} / \sqrt{C_h}} - \frac{1}{N} \frac{\sum W_h p_h q_h}{\sum W_h \sqrt{p_h q_h} / \sqrt{C_h}}$$

$$= \frac{\sum W_h^2 \sqrt{p_h q_h}}{n W_h \sqrt{p_h q_h} / \sqrt{C_h}} \times \frac{\sum W_h \sqrt{p_h q_h} / \sqrt{C_h} - 1}{N} \sum W_h p_h q_h$$

$$V(pst)_{opt.c} = \frac{1}{n} (\sum W_h \sqrt{p_h q_h} C_h) (\sum W_h \sqrt{p_h q_h} / \sqrt{C_h}) - \frac{1}{N} \sum W_h p_h q_h$$

$$V(pst)_{opt.c} = \frac{1}{n} (\sum W_h \sqrt{p_h q_h} C_h) (\sum W_h \sqrt{p_h q_h} / \sqrt{C_h}) - \frac{1}{N} \sum W_h p_h q_h$$

Theorem:- Show that

$$V(pst)_{opt} < V(pst)_{prop} < V(pst)_{sum}$$

Ignoring f.p.c when Sampling is from The large population.

proof:-

$$\text{Let } V(pst)_{sum} = \frac{S_p^2}{n}$$

$$V(pst)_{prop} = \frac{1}{n} \sum W_h p_h q_h \text{ (Ignoring f.p.c)} \quad \text{--- (1)}$$

$$V(pst)_{opt} = \frac{1}{N} \left(\sum_n w_n \sqrt{p_n Q_n} \right)^2 \text{ (ignoring f.p.c.)} \quad (2)$$

By definition

$$\begin{aligned} S_p^2 &= \frac{1}{N-1} \sum_{h=1}^L \sum_{i=1}^{N_h} (p_{hi} - \bar{p})^2 \\ &= \frac{1}{N-1} \sum_{h=1}^L \sum_{i=1}^{N_h} [(p_{hi} - p_h) + (p_h - \bar{p})]^2 \\ &= \frac{1}{N-1} \sum_{h=1}^L \sum_{i=1}^{N_h} [(p_{hi} - p_h)^2 + (p_h - \bar{p})^2 + 2(p_{hi} - p_h)(p_h - \bar{p})] \\ &= \frac{1}{N-1} \left[\sum_{h=1}^L \sum_{i=1}^{N_h} (p_{hi} - p_h)^2 + \sum_{h=1}^L N_h (p_h - \bar{p})^2 + 2 \sum_{h=1}^L \sum_{i=1}^{N_h} (p_{hi} - p_h)(p_h - \bar{p}) \right] \end{aligned}$$

The Crossproduct term vanish because the units drawn from each stratum are independent.

$$\begin{aligned} S_p^2 &= \frac{1}{N-1} \left[\sum_{h=1}^L \sum_{i=1}^{N_h} (p_{hi} - p_h)^2 + \sum_{h=1}^L N_h (p_h - \bar{p})^2 \right] \\ &= \frac{1}{N-1} \left[\sum_{h=1}^L (N_h - 1) S_{p_h}^2 + \sum_{h=1}^L N_h (p_h - \bar{p})^2 \right] \end{aligned}$$

As we known that

$$S_{ph}^2 = \frac{N_h P_h Q_h}{N_h - 1}$$

$$(I_{np} = S_h^2)$$

Since pop is large so.

$$N_h - 1 \cong N_h \quad \text{and} \quad N - 1 \cong N$$

$$S_{ph}^2 = \frac{N_h P_h Q_h}{N_h} = P_h Q_h$$

Now

$$\begin{aligned} S_p^2 &= \frac{1}{N} \left[\sum_{h=1}^L N_h P_h Q_h + \sum_{h=1}^L N_h (P_h - \bar{P})^2 \right] \\ &= \sum_{h=1}^L w_h P_h Q_h + \sum_{h=1}^L w_h (P_h - \bar{P})^2 \end{aligned}$$

Dividing on both sides by N we get.

$$\frac{S_p^2}{n} = \frac{1}{n} \left(\sum_{h=1}^L w_h P_h Q_h \right) + \frac{1}{n} \sum_{h=1}^L w_h (P_h - \bar{P})^2$$

$$V(pst)_{prop} = V(pst)_{bwn} + \frac{1}{n} \sum_{h=1}^L w_h (P_h - \bar{P})^2$$

It shows that prop

$$V(pst)_{prop} < V(pst)_{bwn} \quad \text{--- (A)}$$

In order to show that $V(pst)_{opt} < V(pst)_{prop}$

we subtract eq (5) from eq (1)

$$V(pst)_{prop} - V(pst)_{opt} = \frac{1}{n} (\sum w_n p_n Q_n) - \frac{1}{n} (\sum w_n \sqrt{p_n Q_n})^2$$

$$= \frac{1}{n} [\sum w_n p_n Q_n - (\sum w_n \sqrt{p_n Q_n})^2]$$

Adding and subtracting $(\sum w_n \sqrt{p_n Q_n})^2$ on the R.H.S

$$V(pst)_{prop} - V(pst)_{opt} = \frac{1}{n} [\sum w_n p_n Q_n - 2(\sum w_n \sqrt{p_n Q_n})^2 + (\sum w_n \sqrt{p_n Q_n})^2]$$

$$= \frac{1}{n} [\sum w_n p_n Q_n - 2(\sum w_n \sqrt{p_n Q_n})(\sum w_n \sqrt{p_n Q_n}) + (\sum w_n \sqrt{p_n Q_n})^2]$$

$$\therefore \sum w_n = 1$$

$$= \frac{1}{n} [\sum w_n \sum p_n Q_n - 2(\sum \sqrt{p_n Q_n})(\sum w_n \sqrt{p_n Q_n}) + (\sum w_n \sqrt{p_n Q_n})^2]$$

$$V(pst)_{prop} - V(pst)_{opt} = \frac{1}{n} \sum W_h^2 (P_h S_h - E W_h P_h S_h)^2$$

we see that difference is a positive quantity. So

$$V(pst)_{prop} > V(pst)_{opt}$$

i.e.

$$V(pst)_{prop} = V(pst)_{opt} + \frac{1}{n} \sum W_h^2 (P_h S_h - E W_h P_h S_h)^2$$

we can see

$$V(pst)_{opt} < V(pst)_{prop} \quad \text{--- (B)}$$

Comparing (A) and (B) we get

$$V(pst)_{opt} < V(pst)_{prop} < V(pst)_{surn}$$

Hence proved.

Q: 5.11

If $V(\bar{y}_{st})$ is the variance of the estimated mean from the stratified random sampling of size "n"

with proportional allocation and $V(\bar{y})$ is the variance of the mean from S-R sample of size "n". Then show that

(i) The ratio $\frac{V(\bar{y})_{\text{prop}}}{V(\bar{y})_{\text{st}}}$ does not

depends upon the sampling size but

(ii) The ratio $\frac{V(\bar{y})_{\text{opt}}}{V(\bar{y})_{\text{prop}}}$ decreases

as "n" (sampling size) increases.

Solution:-

$$\text{Let } V(\bar{y})_{\text{sam}} = \frac{N-n}{N \cdot n} S^2$$

$$V(\bar{y})_{\text{prop}} = \frac{N-n}{N \cdot n} \sum_{h=1}^L W_h S_h^2$$

Ratio of these two variances is

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{ram}} = \frac{\frac{N-n}{N \cdot n} \sum_{h=1}^L W_h S_h^2}{\frac{N-n}{N \cdot n} S^2} = \frac{\sum_{h=1}^L W_h S_h^2}{S^2}$$

So, it does not depend on the Sample Size "n"

$$(ii) \quad V(\bar{y}_{st}) = \frac{1}{n} \left(\sum W_h S_h \right)^2 - \frac{1}{N} \sum W_h S_h^2$$

$$V(\bar{y}_{st}) = \frac{N-n}{N \cdot n} \sum_{h=1}^L W_h S_h^2$$

$$= \left(\frac{1}{n} - \frac{1}{N} \right) \sum W_h S_h^2$$

$$= \frac{1}{n} \sum W_h S_h^2 - \frac{1}{N} \sum W_h S_h^2$$

Hence The ratio

$$\frac{V(\bar{y}_{st})_{opt}}{V(\bar{y}_{st})_{prop}} = \frac{\frac{1}{n} \left(\sum W_h S_h \right)^2 - \frac{1}{N} \sum W_h S_h^2}{\frac{1}{n} \sum W_h S_h^2 - \frac{1}{N} \sum W_h S_h^2}$$

$$\frac{V(\bar{y}_{st})_{opt}}{V(\bar{y}_{st})_{prop}} = \frac{\frac{1}{n} \left(\sum W_h S_h \right)^2 - \frac{1}{N} \sum W_h S_h^2}{\frac{1}{n} \sum W_h S_h^2 - \frac{1}{N} \sum W_h S_h^2}$$

We see that the same quantity

is being subtracted from the numerator and denominator which is also known as f.p.c and we also know that

$$\sum W_h S_h^2 > (\sum W_h S_h)^2$$

So, it is a decreasing function as the whole equation depends on the Sample Size "n"

As "n" will increase the function (ratio) will decrease.

Q: 5.9

with two strata, a Sampler would like to have $\mu_1 = \mu_2$ for administrative convenience, instead of using the values given by the Neyman allocation. If $V(\bar{y}_{st})$, $V_{opt}(\bar{y}_{st})$ denote the variance by the given $n_1 = n_2$ and the Neyman allocations, respectively, show that the fractional increase in variance

$$\frac{V(\bar{y}_{st}) - V_{opt}(\bar{y}_{st})}{V_{opt}(\bar{y}_{st})} = \left(\frac{r-1}{r+1} \right)^2$$

where $h = \frac{n_1}{n_2}$ as given by

Neyman allocation and
f.p.c is ignored

proof:-

$$V(\bar{y}_{st}) = \sum_{h=1}^L \frac{w_h^2 S_h^2}{n_h} (1 - f_h)$$

$$= \sum_{h=1}^L \frac{w_h^2 S_h^2}{n_h}$$

$$= \frac{1}{n} \sum_{h=1}^L w_h^2 S_h^2$$

$$= \frac{1}{n} \sum_{h=1}^L w_h^2 S_h^2$$

For two strata we have

$$V(\bar{y}_{st}) = \frac{1}{n} [w_1^2 S_1^2 + w_2^2 S_2^2]$$

$$V(\bar{y}_{st})_{opt} = \frac{1}{n} \left[\frac{(\sum w_h S_h)^2}{N} - \frac{1}{N} \sum w_h^2 S_h^2 \right]$$

$$= \frac{1}{n} (\sum w_h S_h)^2 \quad (\text{ignoring f.p.c.})$$

$$= \frac{1}{n} [w_1 S_1 + w_2 S_2]^2$$

If f.p.c is
ignored place
with two strata

$$n = n_1 + n_2$$

and $n_1 = n_2$

$$n = n_1 + n_2$$

$$n = n_1 + n_2$$

$$n_1 = \frac{n}{2}$$

Similarly

$$n_2 = \frac{n}{2}$$

Now,

$$\begin{aligned}
 \frac{V(\bar{y}_{st}) - V(\bar{y}_{st})_{opt}}{V(\bar{y}_{st})_{opt}} &= \frac{\frac{2}{n} [w_1^2 S_1^2 + w_2^2 S_2^2] - \frac{1}{n} [w_1 S_1 + w_2 S_2]^2}{\frac{1}{n} [w_1 S_1 + w_2 S_2]^2} \\
 &= \frac{2w_1^2 S_1^2 + 2w_2^2 S_2^2 - w_1^2 S_1^2 - w_2^2 S_2^2 - 2w_1 S_1 w_2 S_2}{[w_1 S_1 + w_2 S_2]^2} \\
 &= \frac{w_1^2 S_1^2 + w_2^2 S_2^2 - 2w_1 S_1 w_2 S_2}{[w_1 S_1 + w_2 S_2]^2} \\
 &= \frac{[w_1 S_1 - w_2 S_2]^2}{[w_1 S_1 + w_2 S_2]^2}
 \end{aligned}$$

Dividing the numerator and denominator by $w_2^2 S_2^2$ on the R.H.S

$$\frac{V(\bar{y}_{st}) - V(\bar{y}_{st})_{opt}}{V(\bar{y}_{st})_{opt}} = \frac{\left[\frac{w_1 S_1}{w_2 S_2} - 1 \right]^2}{\left[\frac{w_1 S_1}{w_2 S_2} + 1 \right]^2} = \left(\frac{r-1}{r+1} \right)^2$$

As $r = \frac{n_1}{n_2}$ by Neyman allocation

$$n_1 = \frac{n W_1 S_1}{W_1 S_1 + W_2 S_2} \quad , \quad n_2 = \frac{n W_2 S_2}{W_1 S_1 + W_2 S_2}$$

$$\frac{n_1}{n_2} = \frac{n W_1 S_1}{W_1 S_1 + W_2 S_2} \div \frac{n W_2 S_2}{W_1 S_1 + W_2 S_2}$$

$$= \frac{W_1 S_1}{W_1 S_1 + W_2 S_2} \times \frac{W_1 S_1 + W_2 S_2}{W_2 S_2}$$

$$= \frac{W_1 S_1}{W_2 S_2}$$

But given $\frac{n_1}{n_2} = \gamma$ so $\frac{W_1 S_1}{W_2 S_2} = \gamma$

$$\frac{V(\bar{y}_{st}) - V(\bar{y}_{st})_{opt}}{V(\bar{y}_{st})_{opt}} = \frac{(\gamma - 1)^2}{(\gamma + 1)^2} = \left(\frac{\gamma - 1}{\gamma + 1} \right)^2$$

Q No. 5.5

A Sampler has two strata with relative sizes W_1, W_2 . He believes that S_1, S_2 can be taken as equal but thinks that C_2 may be between $2C_1$ and

4C, - He would prefer to use proportional allocation but does not wish to incur a substantial increase in variance compared with optimum allocation. For a given cost $C = C_1 n_1 + C_2 n_2$ ignoring the fpc show

$$\frac{v(\bar{y}_{st})_{\text{prop}}}{v(\bar{y}_{st})_{\text{opt}}} = \frac{w_1 C_1 + w_2 C_2}{(w_1 \sqrt{C_1} + w_2 \sqrt{C_2})^2}$$

If $w_1 = w_2$ Compute the relative increases in variance from using proportional allocation when $C_1/C_2 = 2, 4$.

Solution:-

The table of information

is

n_h	n	w_h	S_h	C_h
n_1	1	w_1	S_1	C_1
n_2	2	w_2	S_2	$2C_1 \leq C_2 \leq 4C_1$

Let

$$V(\bar{y}_{st})_{\text{prop}} = \frac{1}{n} \sum w_h S_h^2 \quad \text{Ignoring f.p.c.}$$
$$= \frac{1}{n} [w_1 S_1^2 + w_2 S_2^2]$$

As

$$S_1^2 = S_2^2 = S^2$$

So,

$$V(\bar{y}_{st})_{\text{prop}} = \frac{S^2}{n} [w_1 + w_2] \quad \text{--- (A)}$$

As optimum allocation with fixed cost is

$$n_h = n \frac{w_h S_h}{\sum w_h S_h}$$

Then

$$n_1 = \frac{n w_1 S_1}{w_1 S_1 + w_2 S_2}$$

$$\text{As } S_1 = S_2 = S$$

So,

$$n_1 = n \frac{w_1}{w_1 + w_2}, \quad n_2 = n \frac{w_2}{w_1 + w_2}$$

Substituting the value of n_1 and n_2 in the cost function we will have.

$$C = C_1 n_1 + C_2 n_2$$

$$= C_1 \frac{n w_1}{w_1 + w_2} + C_2 \frac{n w_2}{w_1 + w_2}$$

$$= \frac{n}{w_1 + w_2} [C_1 w_1 + C_2 w_2]$$

$$\frac{w_1 + w_2}{n} = \frac{C_1 w_1 + C_2 w_2}{C}$$

Substituting this in eq. (A)

$$V(\bar{y}_{st})_{\text{prop}} = S^2 \frac{C_1 w_1 + C_2 w_2}{C}$$

$$= \frac{S^2}{C} [C_1 w_1 + C_2 w_2]$$

$$V(\bar{y}_{st})_{\text{opt. C}} = \frac{1}{n} (\sum w_h s_h \sqrt{C_h}) (\sum w_h s_h / \sqrt{C_h}) \quad \text{ignoring f.p.c.}$$

$$= \frac{1}{n} (w_1 s_1 \sqrt{C_1} + w_2 s_2 \sqrt{C_2}) (w_1 s_1 / \sqrt{C_1} + w_2 s_2 / \sqrt{C_2})$$

$$\text{AS } S_1 = S_2 = S$$

So,

$$V(\bar{y}_{st})_{\text{opt. C}} = \frac{S^2}{n} [(w_1 \sqrt{C_1} + w_2 \sqrt{C_2}) (w_1 / \sqrt{C_1} + w_2 / \sqrt{C_2})] \quad \text{--- (B)}$$

Let the optimum allocation for

Varying cost is given by the following solution.

$$n_1 = n \frac{w_1 S_1 / \sqrt{C_1}}{\frac{w_1 S_1}{\sqrt{C_1}} + \frac{w_2 S_2}{\sqrt{C_2}}}$$

$$n_1 = n \frac{w_1 S_1 / \sqrt{C_1}}{\frac{w_1 S_1}{\sqrt{C_1}} + \frac{w_2 S_2}{\sqrt{C_2}}}$$

$$n_2 = n \frac{w_2 S_2 / \sqrt{C_2}}{\frac{w_1 S_1}{\sqrt{C_1}} + \frac{w_2 S_2}{\sqrt{C_2}}}$$

Substituting the value of n_1 and n_2 in to the cost function Hence the cost function is $C = C_1 n_1 + C_2 n_2$

$$C = \frac{\sqrt{C_1}}{\frac{w_1 S_1}{\sqrt{C_1}} + \frac{w_2 S_2}{\sqrt{C_2}}} n \frac{w_1 S_1 / \sqrt{C_1}}{\frac{w_1 S_1}{\sqrt{C_1}} + \frac{w_2 S_2}{\sqrt{C_2}}} + \frac{\sqrt{C_2}}{\frac{w_1 S_1}{\sqrt{C_1}} + \frac{w_2 S_2}{\sqrt{C_2}}} n \frac{w_2 S_2 / \sqrt{C_2}}{\frac{w_1 S_1}{\sqrt{C_1}} + \frac{w_2 S_2}{\sqrt{C_2}}}$$

$$= \frac{n [\sqrt{C_1} w_1 S_1 + \sqrt{C_2} w_2 S_2]}{\frac{w_1 S_1}{\sqrt{C_1}} + \frac{w_2 S_2}{\sqrt{C_2}}}$$

$$\frac{C}{n} = \frac{\sqrt{C_1} w_1 S_1 + \sqrt{C_2} w_2 S_2}{\frac{w_1 S_1}{\sqrt{C_1}} + \frac{w_2 S_2}{\sqrt{C_2}}}$$

$$\text{AS } S_1 = S_2 = S$$

$$C = \frac{w_1 \sqrt{C_1} + w_2 \sqrt{C_2}}{n}$$

$$n = \frac{w_1}{\sqrt{C_1}} + \frac{w_2}{\sqrt{C_2}}$$

$$\frac{L}{n} = \frac{L}{C} \left[\frac{w_1 \sqrt{C_1} + w_2 \sqrt{C_2}}{\frac{w_1}{\sqrt{C_1}} + \frac{w_2}{\sqrt{C_2}}} \right]$$

putting this in eq (B)

$$\begin{aligned} V(\bar{y}_{st}) &= S^2 \left[(w_1 \sqrt{C_1} + w_2 \sqrt{C_2}) \left(\frac{w_1}{\sqrt{C_1}} + \frac{w_2}{\sqrt{C_2}} \right) \right] \times \frac{1}{C} \left[\frac{w_1 \sqrt{C_1} + w_2 \sqrt{C_2}}{\frac{w_1}{\sqrt{C_1}} + \frac{w_2}{\sqrt{C_2}}} \right] \\ &= \frac{S^2 (w_1 \sqrt{C_1} + w_2 \sqrt{C_2})^2}{C} \end{aligned}$$

Now,

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}}$$

$$= \frac{S^2 [C_1 w_1 + C_2 w_2]}{S^2 [w_1 \sqrt{C_1} + w_2 \sqrt{C_2}]^2}$$

$$= \frac{C_1 w_1 + C_2 w_2}{(w_1 \sqrt{C_1} + w_2 \sqrt{C_2})^2} \quad \text{--- (C)}$$

which is proved

Now for next given

$w_1 = w_2 = w$ say $C_2/C_1 = 2$ and

$C_2/C_1 = 4$

Put $w_1 = w_2 = w$ in eq (C)

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}} = \frac{(wC_1 + wC_2)}{(w\sqrt{C_1} + w\sqrt{C_2})^2} = \frac{w(C_1 + C_2)}{w^2(\sqrt{C_1} + \sqrt{C_2})^2}$$

$$= \frac{C_1 + C_2}{w(C_1 + C_2 + 2\sqrt{C_1C_2})}$$

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}} = \frac{C_1 + C_2}{\frac{1}{2}(C_1 + C_2 + 2\sqrt{C_1C_2})}$$

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}} = \frac{C_1 + C_2}{\frac{1}{2}(C_1 + C_2 + 2\sqrt{C_1C_2})}$$

$$\begin{aligned} w_1 &= w_1 + w_2 \\ 1 &= w + w \\ 1 &= 2w \\ w &= \frac{1}{2} \end{aligned}$$

Dividing numerator and denominator by (C_1)

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}} = \frac{2(1 + C_2/C_1)}{(1 + C_2/C_1 + 2\sqrt{C_2/C_1})} \quad \text{--- (D)}$$

Now put $C_2/C_1 = 2$ in eq (D)

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}} = \frac{2(1+2)}{(1+2+2\sqrt{2})} = \frac{6}{3+2\sqrt{2}}$$

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}} = 1.0294$$

So increase in case $C_2/C_1 = 2$ is 0.0294

now put $C_2/C_1 = 4$

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}} = \frac{2(1+4)}{(1+4+2(\sqrt{4}))}$$

$$= \frac{10}{9}$$

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}} = 1.11$$

$$\frac{V(\bar{y}_{st})_{prop}}{V(\bar{y}_{st})_{opt}}$$

So increase in case $C_2/C_1 = 4$ is 0.11